

Problem 1:

- Run a container nginx with name my-nginx and attach 2 volumes to the container using volume mount
- Volume1 for containing static html file
- Volume2 for containing nginx configuration
- Edit the html content
- Remove the container
- Run a new 2 containers with the following:
 - Attach the two volumes that were attached to the previous container using volume mount
 - Map port 80 to port 8080 on you host machine
 - Access the html files from your browser

```
C:\Users\basil>docker volume create static
static
```

```
C:\Users\basil>docker volume create configs
configs
```

```
C:\Users\basil>docker run -d --name my-nginx -v static:/usr/share/nginx/html -v config:/etc/nginx/config.d nginx
2b1c472ab740e70cb864fa9b4004a76730773b5f3d5dc1dc2aeb86e2c82798f9

C:\Users\basil>docker exec my-nginx sh -c "echo 'hello world'> /usr/share/nginx/html/index.html"

C:\Users\basil>docker exec my-nginx cat /usr/share/nginx/html/html.index
cat: /usr/share/nginx/html/html.index: No such file or directory

C:\Users\basil>docker exec my-nginx cat /usr/share/nginx/html/index.html
hello world
```

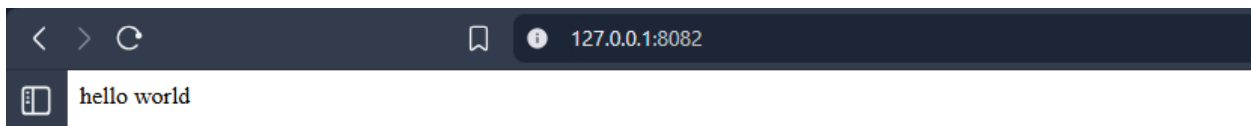
```
C:\Users\basil>docker stop my-nginx
my-nginx
```

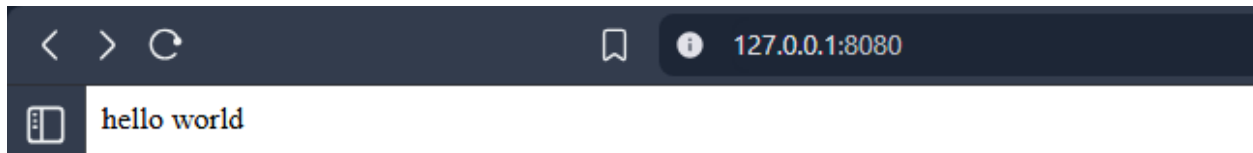
```
C:\Users\basil>docker rm my-nginx
my-nginx
```

```
C:\Users\basil>
C:\Users\basil>docker run -d --name nginx_C1 -p 8080:80 -v static:/usr/share/nginx/html -v configs:/etc/nginx/conf.d nginx
a7fa23e7dfcfe9d2694ea28a27b38368ce36f73a6701740c8270b33a4fe0dd0

C:\Users\basil>docker run -d --name nginx_C2 -p 8081:80 -v html:/usr/share/nginx/html -v configs:/etc/nginx/conf.d nginx
f8d0f4b7fbed8709103bdd13cb7bad48e078dbe74ff6999e2ac9f48f499701ad

C:\Users\basil>docker run -d --name nginx_C3 -p 8082:80 -v static:/usr/share/nginx/html -v configs:/etc/nginx/conf.d nginx
c9cbe33f61beba7b73d4dfb1ecf6607191793039318876f563a553918ffc60c6
```





Problem 2:

- Run a container Nginx with name nginx-bind-mount and attach 2 volumes using bind mount under any paths
- Remove the container
- Run a new container with the following:
 - Attach the two volumes that were attached to the previous container
 - Check the old data in the new containers

```
C:\Users\basil>mkdir html_bind
C:\Users\basil>mkdir config_bind
C:\Users\basil>echo this is bind mounting > html_bind/index.html
```

```
C:\Users\basil>docker run -d --name nginx_bind -p 8080:80 -v "%cd%\html_bind":/usr/share/nginx/html -v "%cd%\config_bind":/etc/nginx/conf.d nginx
ffbc7814cebdd0fbb18977bb49e54fd45039374938bcdea114b54085c74be0cd
```

```
C:\Users\basil>docker stop nginx_bind
nginx_bind

C:\Users\basil>docker rm nginx_bind
nginx_bind
```

```
C:\Users\basil>docker run -d --name nginx_new_bind -p 8081:80 -v "%cd%\html_bind":/usr/share/nginx/html -v "%cd%\conf-bind":/etc/nginx/conf.d nginx
efcc9304d48a77d9d2f24fa6b3b0809032ccd83b62e29eab9b24593748ee15f4
```

```
C:\Users\basil>docker run -d --name nginx_new_bind2 -p 8083:80 -v "%cd%\html_bind":/usr/share/nginx/html -v "%cd%\config_bind":/etc/nginx/conf.d nginx
e874e5c3f93948833170c64709ec0436086931496ebf973cf3cbc1d3edabb6fb

C:\Users\basil>docker exec nginx_new_bind2 cat /usr/share/nginx/html/index.html
this is bind mounting
```

Problem 3:

- Create 2 nginx containers with 2 different network of type bridge, enter to one of them and use curl command to view the content of the other container.

```
C:\Users\basil>docker network create -d bridge A
91c999c922d36d4e98c9bc5e9d0b6badd35fac75a3ede98fae784e5088107003
```

```
C:\Users\basil>docker network create -d bridge B
7241cd3360009b81abb36b663f5475f04214456cf3df1aaa2b5b21bfbb1be26e
```

```
C:\Users\basil>docker run -d --name nginx_A --network A nginx
93d56bf27c8c1eb438b396b8e994d1d6f8c123bfaeb262a5e6ea8a17557edecb
```

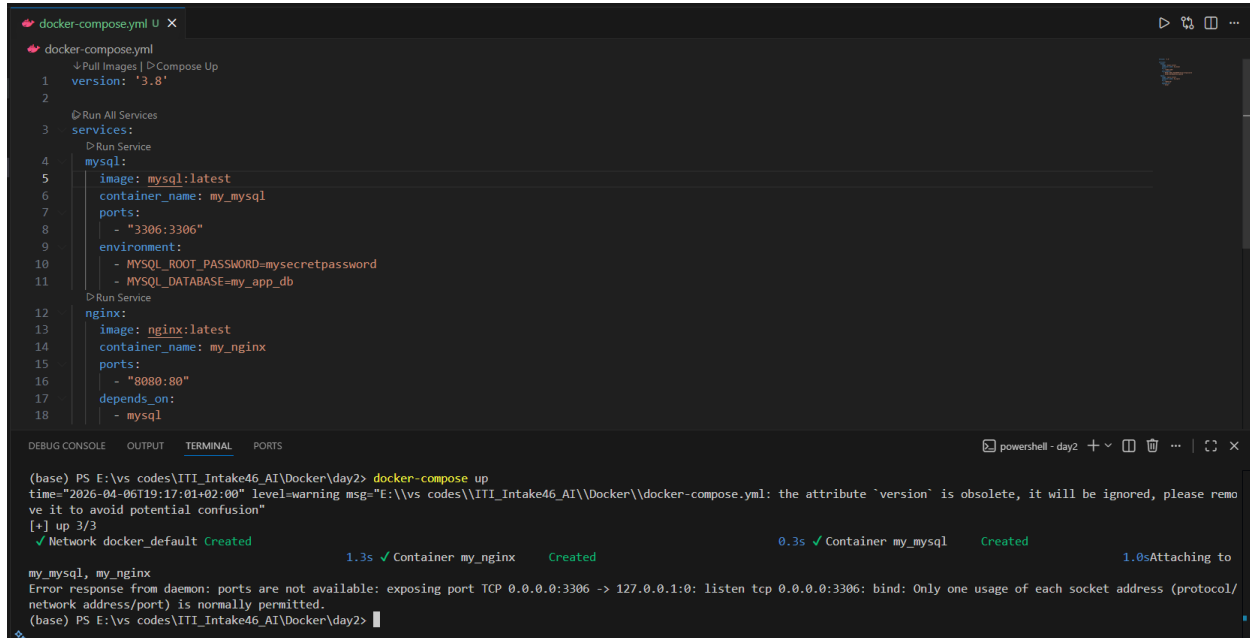
```
C:\Users\basil>docker run -d --name nginx_B --network B nginx
46bb5019cb53d66e3dc0adcf8a9469af8ad3d4bb667d147d0a277fd7098126af
```

- Because they are isolated, I can't enter the other container

```
C:\Users\basil>docker exec -it nginx_A bash
root@93d56bf27c8c:/# curl nginx_B
curl: (6) Could not resolve host: nginx_B
root@93d56bf27c8c:/#
```

Problem 4:

- Create docker compose with:
 - Two services nginx and mysql
 - Add needed ports and environments for both services
 - nginx service is depending on mysql service



```
docker-compose.yml
1 version: '3.8'
2
3 services:
4   mysql:
5     image: mysql:latest
6     container_name: my_mysql
7     ports:
8       - "3306:3306"
9     environment:
10      - MYSQL_ROOT_PASSWORD=mysecretpassword
11      - MYSQL_DATABASE=my_app_db
12   nginx:
13     image: nginx:latest
14     container_name: my_nginx
15     ports:
16       - "8080:80"
17     depends_on:
18       - mysql

(base) PS E:\vs codes\ITI_Intake46_AI\Docker\day2> docker-compose up
time="2026-04-06T19:17:01+02:00" level=warning msg="E:\vs codes\ITI_Intake46_AI\Docker\day2\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 3/3
  ✓ Network docker_default Created
1.3s ✓ Container my_nginx Created
0.3s ✓ Container my_mysql Created
1.0s Attaching to
my_mysql, my_nginx
Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:3306 -> 127.0.0.1:0: listen tcp 0.0.0.0:3306: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.
(base) PS E:\vs codes\ITI_Intake46_AI\Docker\day2>
```